$$x(t) = u(t-2) * (tu(t))$$

convolution property.

$$\mathcal{L}\{x_1(t) * x_2(t)\} = X_1(s) \cdot X_2(s)$$

$$\mathcal{L}\{u(t-2)\} = \frac{e^{-2s}}{s}$$

$$\mathcal{L}\{t^n u(t)\} = \frac{n!}{s^{n+1}} \; ; \quad \mathcal{L}\{t u(t)\} = \frac{1}{s^2}$$

$$\mathcal{L}\{u(t-2) * t^n u(t)\} = \frac{e^{-2s}}{s} \cdot \frac{1}{s^2}$$

$$= \frac{e^{-2s}}{s^3}$$